Defensive Publication

Neo Para Q-Axiom™

A Deterministic Integrity Axiom for Governance-Critical Systems

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1. Abstract

This publication discloses a deterministic integrity system, herein referred to as **Neo Para Q-Axiom™**, designed for governance-critical systems where execution, decision-making, or action must be halted unless all asserted facts can be traced to verified source material.

The system introduces a **binary, fail-closed integrity axiom** that operates independently of policy, probability scoring, optimization, or subjective judgment. Its purpose is to prevent narrative contamination, evidentiary drift, and unverified inference in high-risk domains such as legal, regulatory, compliance, safety-critical AI, and automated decision systems.

This disclosure is intended to establish prior art and prevent exclusive claims over the disclosed concepts.

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2. Technical Field

The disclosed system relates to:

- governance and assurance systems,
- integrity controls,
- provenance verification,
- evidence-based decision systems,
- AI and automated reasoning oversight,
- compliance and audit architectures.

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3. Problem Statement

Existing governance, AI safety, and compliance systems commonly rely on:

- probabilistic confidence scores,
- advisory warnings,
- post-hoc review,

- policy-based moderation.

These approaches allow systems to proceed despite:

- unverified sources,
- inferred meaning beyond evidence,
- pattern inflation,
- reference errors,
- gradual narrative drift.

In governance-critical contexts, such permissive behavior creates unacceptable risk.

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4. Core Principle: Neo Para Q-Axiom™

Neo Para Q-Axiom™ is defined as follows:

System execution must not proceed unless every asserted fact can be traced to verified source material without inference, inflation, or ambiguity.

Key properties:

- **Deterministic:** outcomes are binary (PASS / FAIL)
- **Fail-closed:** failure halts execution
- **Pre-execution:** applied before action, filing, or deployment
- **Non-probabilistic:** no scoring or confidence thresholds
- **Policy-agnostic:** operates independently of ethical or policy frameworks

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5. Q-Axiom Integrity Gate™ (Core Mechanism)

The primary enforcement mechanism is the **Q-Axiom Integrity Gate™**.

5.1 Function

The gate performs a mandatory self-audit on constructed facts, outputs, or decisions **before execution**.

If any audit check fails, execution is halted.

5.2 Audit Checks (Enabling Disclosure)

For each asserted fact or output, the following checks are performed:

1. **Evidence Boundary Check**
    - Question: Does any part of the statement introduce meaning, implication, or conclusion not explicitly present in the source material?
    - If yes → **FAIL**

2. **Language Inflation Check**
    - Question: Does the wording imply intent, causality, severity, or pattern not explicitly stated in the source?
    - If yes → **FAIL**
3. **Pattern Threshold Check**
    - Question: Is a "pattern" claimed without multiple, independently verified instances?
    - If yes → **FAIL**
4. **Reference Re-Verification Check**
    - Question: Do all references resolve correctly to the cited source material upon direct inspection?
    - If any mismatch → **FAIL**

5.3 Output

The gate produces only one of two outcomes:

- **PASS —** execution may proceed
- **FAIL —** execution is halted

No partial scores, warnings, or mitigation suggestions are produced at this stage.

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6. Q-Axiom Provenance Spine™

The system requires a single, append-only provenance structure referred to as the **Q-Axiom Provenance Spine™**.

6.1 Properties

- Append-only (facts may not be modified or deleted once recorded)
- No summaries or abstractions
- Each fact contains:
    - source reference(s)
    - verification status
    - audit outcome
- Every fact must be traceable backward to original source material

6.2 Purpose

The Provenance Spine enables:

- drill-down verification,

- audit defensibility,
- chain-of-custody integrity,
- resistance to narrative reconstruction.

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7. Q-Axiom Cascade™ (Nested Verification)

The system supports **recursive, multi-level verification** via an integrity cascade.

7.1 Definition

Verified outputs from one project or domain may be treated as axiomatic inputs to a higher-level project **only if**:

- the original provenance remains intact,
- traceability to original sources is preserved,
- all Integrity Gate checks are re-applied at the higher level.

7.2 Effect

This allows:

- cross-domain verification,
- complexity reduction without information overload,
- hierarchical governance analysis without loss of integrity.

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8. Intended Applications (Non-Exhaustive)

- Legal and regulatory analysis
- Compliance automation
- AI agent execution control
- Safety-critical reporting
- Audit and assurance systems
- Decision systems with real-world consequences

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9. Distinguishing Characteristics

The disclosed system is distinguished by:

- binary enforcement (no probabilistic tolerance),
- pre-execution halting behavior,
- explicit provenance requirements,

- independence from policy frameworks,
- applicability across human and AI systems.

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10. Prior Art Declaration

This document is published intentionally to:

- establish public prior art,
- prevent exclusive patent claims over the disclosed concepts,
- preserve freedom to operate for all parties.

No license or ownership over “truth,” “integrity,” or “governance” is asserted by this publication.

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11. Closing Statement

Neo Para Q-Axiom™ defines integrity as a **precondition for action**, not an aspiration.

By making execution contingent on verified provenance, the system converts honesty from a value into a mechanical constraint.